

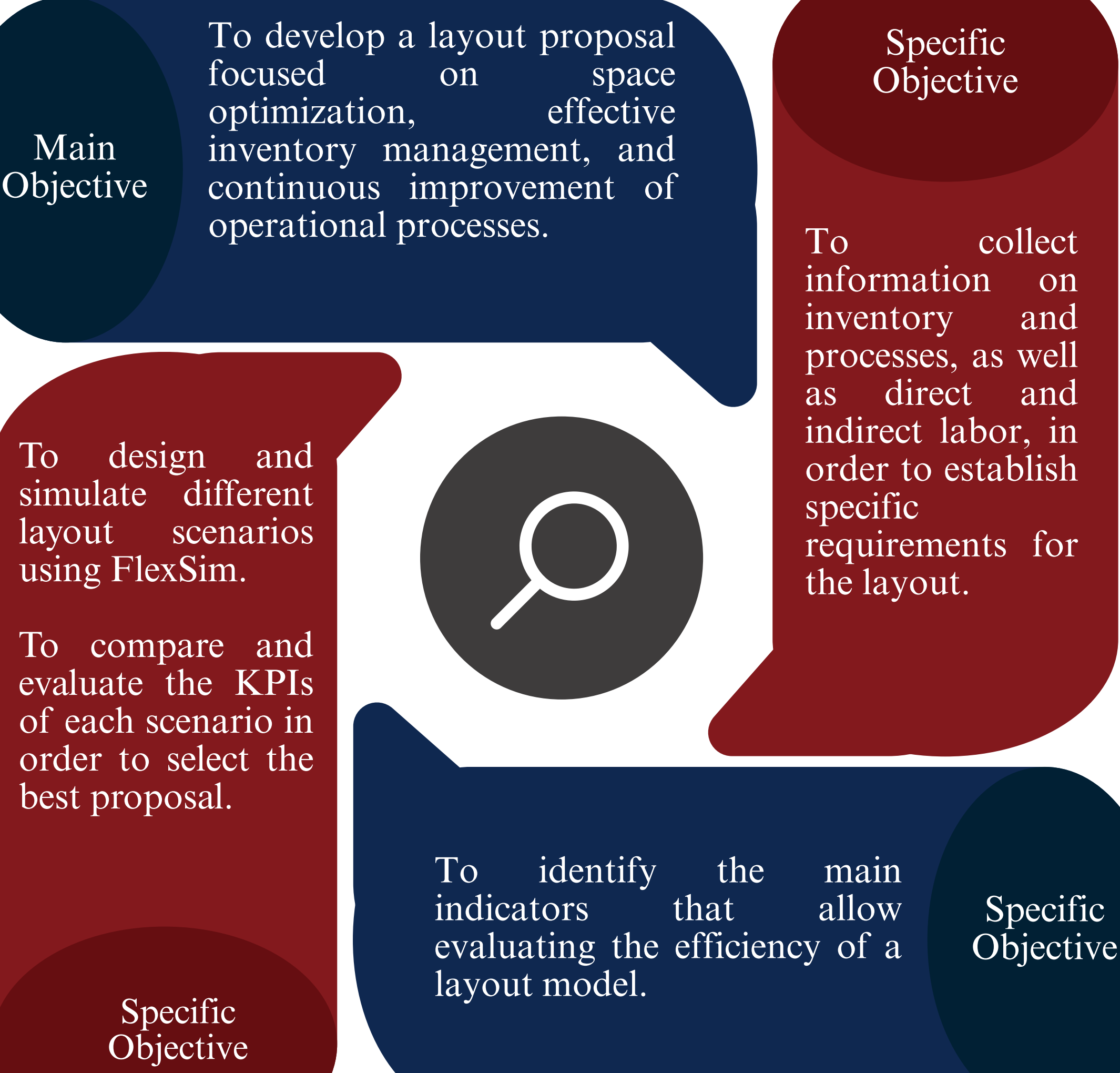
LAYOUT PROPOSAL FOR THE LOGISTICS AND OPERATIONAL SUPPORT CENTER OF ETAPA EP

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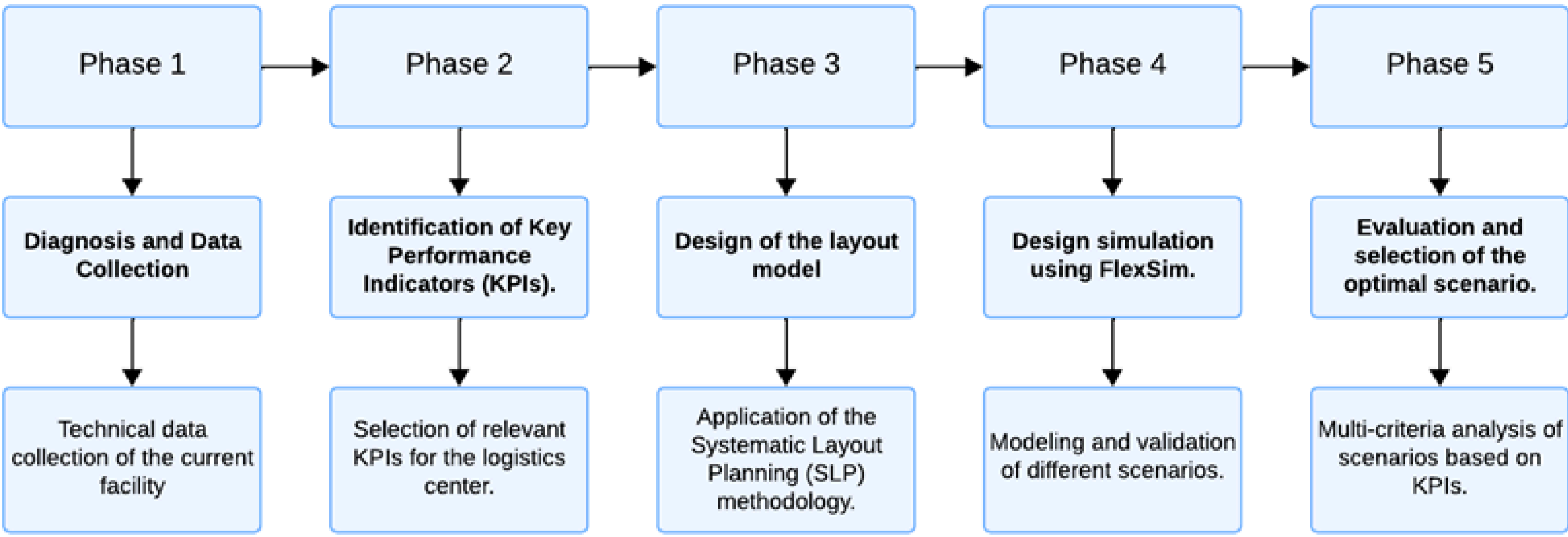
Abstract

To determine and design a layout proposal for the logistics and operations center of ETAPA EP through the use of Industrial Engineering tools and simulation.

Objectives



Methodology



Curricular Integration

Industrial CAD Design.
Logistics and Supply Chain Management.
Process Engineering.
Manufacturing Process Simulation.

Applied Engineering

SLP Methodology.
Application of ABC-XYZ Analysis.
Process Mapping.
Discrete Event Simulation using FlexSim.

Keywords

- Facility layout
- Logistics center
- Simulation
- Optimization

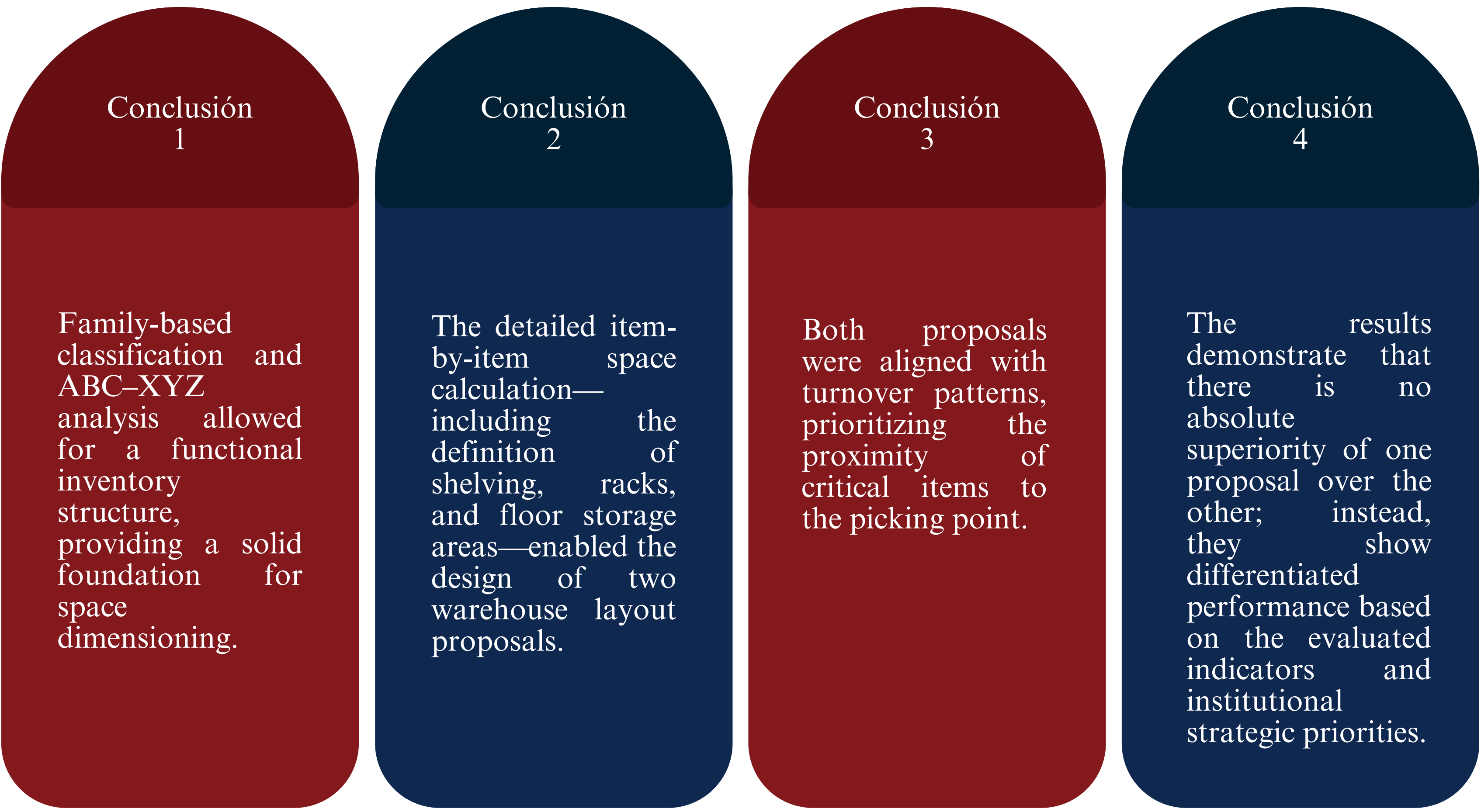
Results

Inventory is highly concentrated: category A (177 items) accounts for 80% of consumption value (USD 295,139.4), while category C (1,209 items) represents only 5% of the value.

The capacity calculation shows a total storage requirement of 5,331.88 m², so vertical storage (4,001.64 m² equivalent) using shelving/racks is essential

In simulation, Proposal 2 improves speed and costs (e.g., total reference cost USD 79.50 vs USD 363.56) but increases physical workload (operators' distance 8,122.5 m vs 5,794.2 m); for sustainability/ergonomics and space utilization (67.43% vs 62.71%), Proposal 1 is recommended as the baseline.

Conclusions



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References

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